Cross-lingual intervention for low-resource Machine Translation

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Abstract

Large Language Models (LLMs) exhibit strong performance across many natural language tasks, yet their capabilities remain highly uneven across languages, particularly in low-resource settings. Existing approaches to mitigate these gaps rely primarily on continued pretraining or fine-tuning , which are computationally expensive and data-intensive. This work studies inference-time cross-lingual intervention (INCLINE), an approach that improves multilingual performance by intervening on internal representations during inference without updating any parameters of the underlying language model. We analyze limitations of the original INCLINE framework, including its focus on discriminative benchmarks, high-resource translation directions, and a mismatch between the alignment objective used during training and the intervention applied at inference time. To address these issues, we propose a consistency-aware alignment objective that directly optimizes the post-intervention representation. We evaluate both approaches in a strictly low-resource machine translation setting across diverse languages, scripts, and tokenization regimes, reporting BLEU, ChrF++, and AfriCOMET. Our results show that inference-time intervention remains effective in severe low-resource scenarios, while the proposed objective improves stability and performance for several African languages. All our code is available on Github[1]

1 Introduction

The rapid progress of large language models has led to unprecedented performance on a wide range of natural language processing tasks. However, this progress has been highly uneven across languages [1]. Most contemporary LLMs are trained predominantly on English or high-resource multilingual data, resulting in representations that generalize poorly to low-resource languages [2]. This imbalance raises both practical and ethical concerns, as large portions of the global population remain underserved by state-of-the-art language technologies.

The dominant approaches to mitigating cross-lingual performance gaps rely on supervised fine-tuning [3], multilingual continued pretraining [4], or parameter-efficient adaptations [5]. While effective, these methods require large parallel corpora, substantial computational resources, and repeated training cycles. In many real-world scenarios, such requirements are prohibitive, motivating the search for alternatives not relying on modifying model parameters.

Inference-time cross-lingual intervention (INCLINE) [6] represents a promising direction in this regard. Instead of altering model weights, INCLINE intervenes directly on internal representations during inference, aligning hidden states of a low-performing language with those of a high-performing language. This enables cross-lingual knowledge transfer while preserving the original model parameters and incurring minimal computational overhead.

Despite its appeal, the original INCLINE framework leaves several important questions unresolved. Its empirical evaluation is dominated by discriminative tasks, while generative tasks such as machine translation receive limited attention. Finally, the alignment objective used during training does not fully reflect the intervention applied at inference time, raising questions about the optimality of the learned transformations.

This project aims to address these gaps by conducting a focused and rigorous study of inference-time cross-lingual intervention in low-resource translation settings and by proposing a principled modification to the alignment objective.

[1] https://github.com/shubhasanket/Inference-Time-Alignment/

2 Inference-Time Cross-Lingual Intervention

INCLINE is based on the hypothesis that internal representations of semantically equivalent sentences across languages can be aligned through linear transformations. Given a low-resource source language and a high-resource target language, typically English, the method learns alignment matrices that map source-language hidden states into the target-language representation space.

The framework consists of two stages. During an offline training phase, during which alignment matrices are learned using a small parallel corpus. During inference, these matrices are used to intervene on the hidden states corresponding to a test input in the source language. Importantly, the underlying model (LLM) parameters remain frozen throughout.

3 Methodology

3.1 Learning Cross-Lingual Alignments

Let $\{(x_i^s, x_i^t)\}_{i=1}^N$ denote a parallel dataset between a source language s and a target language t. For each sentence pair, we extract the hidden state of the last token at layer l of the LLM, denoted by $h_{i,l}^s, h_{i,l}^t \in \mathbb{R}^d$. The original INCLINE approach learns an alignment matrix $W_l \in \mathbb{R}^{d\times d}$ by solving the least-squares problem

$$W_l^* = \arg\min_{W_l} \sum_{i=1}^{N} \|W_l h_{i,l}^s - h_{i,l}^t\|^2.$$

This objective seeks a linear mapping that minimizes the distance between projected source representations and target representations. Note that the solution to this objective can be found in closed form, thus making the training process for all layers essentially a 1-step process.

3.2 Inference-Time Intervention

At inference time, given a source-language test input with hidden state $h_{q,l}^s$, the learned alignment matrix is used to compute a projected representation

$$\hat{h}_{q,l} = W_l^* h_{q,l}^s.$$

An intervention is then applied by combining the original and projected representations:

$$h_{q,l}^{\text{int}} = h_{q,l}^s + \alpha \hat{h}_{q,l},$$

where α is a scalar hyperparameter controlling the intervention strength. In practice, intervention is applied only at the last token to minimize disruption to the generation process.

4 Limitations of the Original Framework

A careful examination of the original INCLINE study reveals several limitations that motivate our work. First, the empirical evaluation is heavily skewed toward discriminative benchmarks, such as XStoryCloze [7], which do not fully reflect the challenges of generative tasks. Second, the translation experiments focus primarily on high-resource languages, despite the stated motivation of improving low-resource performance. Moreover, translation quality is evaluated exclusively using BLEU, a metric that is known to correlate poorly with semantic adequacy for LLM-generated text. [8]

More fundamentally, the alignment objective optimized during training does not correspond exactly to the representation used at inference time. The alignment matrix is trained to approximate a direct projection, while inference relies on a mixed representation that combines the original and projected hidden states. This mismatch suggests that the learned alignment may not be optimal for the actual intervention applied. Hence, in this work, we propose to test a consistency-aware alignment objective.

5 Consistency-Aware Alignment Objective

To address this issue, we propose a modified training objective that directly optimizes the post-intervention representation. Instead of minimizing the distance between $W_l h^s_{i,l}$ and $h^t_{i,l}$, we optimize

$$W_l^* = \arg\min_{W_l} \sum_{i=1}^{N} \|(h^s_{i,l} + \alpha W_l h^s_{i,l}) - h^t_{i,l}\|^2.$$

This objective ensures consistency between the training phase and the inference-time intervention, allowing the alignment matrix to learn the precise shift applied during inference. Another principled addition to the objective is a regularization term. Due to the high dimensionality of the parameter space, we apply ℓ_2 regularization to stabilize training.

6 Experimental Setup

6.1 Model and Data

All experiments are conducted using `LLaMA-3-8B-Instruct` [9] as the base model, quantized to enable feasible computation. We focus exclusively on low-resource machine translation, adopting the nomenclature from NLLB [10] to classify low-resource languages. Alignment matrices are trained using 500 parallel sentence pairs sampled from the FLORES devtest set.

Sentence representations are extracted from the last token of each sentence. This choice reflects the fact that the final hidden state is directly involved in next-token prediction and therefore has a strong influence on generation.

6.2 Evaluation

We compare four approaches:(a) A quantized baseline without intervention,(b) the original INCLINE method, (c) the proposed consistency-aware INCLINE variant, and finally (d) LoRA fine-tuning [cite] trained on the same parallel data. With the first three, we aim to investigate the effectiveness of the INCLINE method itself along with the improvements we proposed. With the LoRA fine-tuning, we aim to make a distinction between the efficacy of an explicit next-token prediction objective against the implicit INCLINE objective. Note that (a) serves as our baseline and (d) serves as our topline.

To evaluate translation quality, we employ BLEU [11] and ChrF++ [12], as well as AfriCOMET [13] for African languages. This combination allows us to capture both surface-level overlap and semantic adequacy, addressing the limitations of BLEU-only evaluation. In particular, following previous work [14], we choose ChrF++ as our primary metric. We do the same for the Serbian (Cyrilic), Pashto, Urdu, Burmese and Lao languages.

7 Results and Analysis

We analyze the behavior of inference-time cross-lingual intervention under genuine low-resource conditions, focusing on performance gains, robustness to script and tokenization mismatch, and the effect of our consistency-aware objective. We first revisit representative results from the original INCLINE study to contextualize our evaluation, and then analyze our low-resource translation experiments.

7.1 Revisiting Existing Experimental Evidence

The main experiments that we rely on from the original work are regarding (a) effect of the hyperparameter α on the downstream task, (b) Variation of accuracy across the choice of number of parallel training examples used to train INCLINE and finally (c) Results related to translation task. These results can be found in the original paper [6] in section 5, Section 5 and Section 4.2 respectively. Briefly, from (a) we decide what values of α we need to try, following (b), we fix the size of our training set to 500. Furthermore, considering the source and target languges considered in (c), we conclude, according to the criteria used in NLLB [10], that none of them are low-resource and hence, we choose only language pairs that satisfy this criteria of being low-resource.

Method	Swahili	Luganda	Somali	Igbo	Zulu
Baseline	31.29	24.57	25.95	28.17	27.96
INCLINE ($\alpha = 0.2$)	32.49	25.62	32.1	25.52	28.05
INCLINE ($\alpha = 0.3$)	30.26	23.46	30.91	25.35	26.99
INCLINE ($\alpha = 0.4$)	30.09	18.79	19.37	25.32	17.44
Improved INCLINE	33.99	26.41	32.99	25.57	27.67
LoRA	69.26	40.38	46.09	41.12	45.71

Table 1: Reporting AfriCOMET for the comparative performance of various methods on low resource Latin script African languages.

7.2 Low-Resource Translation Performance

To answer whether inference-time cross-lingual intervention works well for low-resource languages, our results demonstrate that INCLINE consistently improves translation quality relative to the quantized baseline, even in severe low-resource settings. As detailed in Table 1, these gains are particularly strong for African languages, where the improved AfriCOMET scores indicate enhanced semantic adequacy rather than just superficial lexical overlap. This suggests that aligning internal representations enables the effective transfer of language understanding, even when surface forms differ substantially.

Furthermore, our proposed consistency-aware objective (Improved INCLINE) yields additional performance gains for several of these African languages. For example, Swahili and Somali show noticeable improvements over the standard INCLINE method at various intervention strengths. This supports our hypothesis that directly optimizing the post-intervention representation reduces the misalignment between training and inference objectives, leading to more stable and effective interventions. Note that since $\alpha = 0.2$ worked best in case of INCLINE, we use this same choice for our Improved INCLINE method.

Finally, Table 1 shows that LoRA significantly outperforms both the baseline and INCLINE methods in absolute metrics, reflecting the benefit of directly optimizing token-level likelihood. However, LoRA requires explicit training and additional parameters. In contrast, INCLINE introduces negligible inference overhead and no parameter updates, offering a much quicker and highly efficient alternative to improve the base model when extensive retraining is computationally prohibitive.

7.3 Effect of Script and Tokenization

Source	Baseline	INCLINE ($\alpha = 0.2$)	INCLINE ($\alpha = 0.3$)	INCLINE ($\alpha = 0.4$)	Improved INCLINE	LoRA
Nepali	19.3825	22.54	22.66	20.4258	22.64	50.32
Maithili	17.6783	20.4	20.4	20.3515	18.73	51.54
Bhojpuri	18.0121	21.01	19.75	18.4072	19.56	46.21
Sanskrit	16.7085	23.41	20.5	17.7816	22.04	40.04

Table 2: Performance of different methods on Devnagri script. Note that for Improved INCLINE, we choose $\alpha = 0.2$ since this worked best for the INCLINE method.

Source	Baseline	INCLINE ($\alpha = 0.2$)	INCLINE ($\alpha = 0.3$)	INCLINE ($\alpha = 0.4$)	Improved INCLINE	LoRA
Serbian	21.73	21.73	22.26	21.5476	21.73	60.24
Pashto	19.3808	28.2619	21.7249	21.0785	27.4	44.54
Urdu	14.7835	21.8549	20.5869	19.8412	21.36	50.57
Burmese	17.9395	20.0688	19.97	20.2479	19.27	32.92
Lao	16.3136	18.06	16.6805	16.4184	18.29	32.69

Table 3: Performance of different methods on Cyrilic (Serbian), Arabic (Pashto, Urdu), Mon (Burmese) and Lao script. Again, we choose $\alpha = 0.2$.

To determine whether the performance of INCLINE is dependent on the script of the high-resource language, we evaluated the method across a diverse set of writing systems. Performance improvements are more consistent for languages written in Latin scripts, though non-Latin scripts also benefit. For example, Table 2 demonstrates that INCLINE effectively improves translation quality across Devanagari script languages, such as Nepali and Sanskrit. The method also works well for Arabic scripts, with Table 3 showing substantial gains over the baseline for both Pashto and Urdu.

Language	Baseline	INCLINE ($\alpha = 0.2$)	INCLINE ($\alpha = 0.3$)	INCLINE ($\alpha = 0.4$)
Serbian Latin	18.6	25.72	25.27	25.71
Serbian Cyrlic	17.3795	21.73	22.26	21.5476

Table 4: Performance of the same Serbian language on two different scripts (cyrilic and latin)

To directly isolate the impact of script matching between the source and target languages, we compared translation performance on the Serbian language using both Latin and Cyrillic scripts. Table 4 reveals that the gains are noticeably larger when the scripts of the target and source language match. While both scripts improve over the baseline, Serbian Latin achieves a higher absolute score than Serbian Cyrillic when utilizing INCLINE.

Finally, we investigated whether tokenization density affects the performance of inference-time intervention. Importantly, languages with extremely sparse and long token sequences, such as Lao and Burmese, still show measurable gains. As detailed in Table 3, the method successfully elevates performance for these languages despite the segmentation challenges. This indicates that inference-time intervention can partially compensate for representational deficiencies induced by subword segmentation.

7.4 Effect of Regularization

we evaluated the impact of applying L2 regularization during the training of the alignment matrices under the consistency-aware objective. As shown in Table 5, the inclusion of regularization yields mixed absolute results depending on the specific language. For some languages, such as Pashto and Serbian, the regularized objective clearly outperforms the unregularized version, while for others like Urdu, the unregularized approach achieves a slightly higher score.

However, we observe that L2 regularization is absolutely critical for ensuring stable training across the board. Without this regularization, the optimization process can completely diverge for certain languages. This catastrophic failure is most prominently demonstrated in the case of Sanskrit, where the evaluation score plummets to a near-zero 0.01 without L2, but fully stabilizes at 23.41 when it is applied. Therefore, we conclude that regularization is a necessary addition to mitigate these divergence effects and control overparameterization when learning high-dimensional mappings from limited data.

Source	Baseline	Without L2	With L2
Nepali	22.54	21.46	22.54
Maithili	20.4	18.79	20.4
Bhojpuri	21.01	20.05	21.01
Sanskrit	23.41	0.01	23.41
Serbian	17.38	21.43	22.36
Pashto	19.38	24.63	28.26
Urdu	14.78	25.34	21.86
Burnese	17.94	20.23	20.07
Lao	16.32	17.95	18.06

Table 5: With vs without L2 Regularization. ($\alpha = 0.2$) in both cases

8 Conclusion

This work demonstrates that inference-time cross-lingual intervention (INCLINE) is a viable and highly efficient approach for improving low-resource machine translation without modifying underlying model parameters. By addressing a fundamental inconsistency in the original alignment objective, our proposed consistency-aware method improves overall stability and translation performance, yielding particularly strong gains for several African languages. Our evaluation reveals that while performance improvements are most pronounced when the source and high-resource target languages share a Latin script , the method also produces measurable gains for non-Latin scripts and languages challenged by extremely sparse tokenization, such as Lao and Burmese. Crucially, we found that L2 regularization is essential for the stable training of these alignment matrices, effectively preventing optimization divergence in languages like Sanskrit. Although explicit fine-tuning methods like LoRA remain superior in absolute translation quality , INCLINE introduces negligible inference overhead and requires no parameter updates, providing a compelling performance-efficiency trade-off for rapid adaptation in severely constrained settings.

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